

EXPERIMENT 28: FORWARD CHAINING

Aim

To write a **Prolog program to implement Forward Chaining** using facts and rules and derive new facts from the available knowledge.

Algorithm

1. Start.
2. Define initial facts.
3. Define rules for deriving new facts.
4. Check the available facts against the rules.
5. Add newly derived facts to the knowledge base.
6. Continue until no new facts can be derived.
7. Query the required conclusion.
8. Display the result.
9. Stop.

Source Code

```
% Forward Chaining
```

```
fact(sunny).
```

```
fact(warm).
```

```
rule(sunny, pleasant).
```

```
rule(warm, comfortable).
```

```
rule(pleasant, happy).
```

```
rule(comfortable, happy).
```

```
derive(Fact) :-
```

```
    fact(Fact).
```

derive(Result) :-

rule(Fact, Result),

derive(Fact).

Queries and Output

```
?- [forward_chaining].  
% forward_chaining.pl compiled 0.00 sec, 8 clauses  
  
?- derive(pleasant).  
true.  
  
?- derive(happy).  
true.  
  
?-
```

Result

The Prolog program successfully demonstrates **Forward Chaining** by deriving new conclusions from existing facts and rules.